

An Equivariant-Zeta Audit of Cat-Map Centralizer Quotients

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Abstract

We audit what standard finite equivariant, orbifold, and quotient-stack constructions retain after a cat-map centralizer quotient removes the source dynamics. For a finite abelian group C , translation by $a \in C$ on $X = \coprod_K n_K(C/K)$ has point period $d_K = [\langle a \rangle : \langle a \rangle \cap K]$ and $[C : \langle a \rangle K]$ cycles on each orbit type. The fixed-point rational Burnside zeta retains these source periods but depends on a only through $\langle a \rangle$. The fixed-orbit integral Burnside zeta has support only at period one. Stronger labelled permutation and enhanced carriers retain the translation modulo the action kernel, while their orbifold or Morita quotient images have static dynamics. On the regular centralizer torsor of $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ over $\mathbb{Z}/q\mathbb{Z}$, no one of four scalar-reduction types has source support and unit exponent uniformly over the locked nine-row family. The sole individual exception is point-cardinality reduction at $q = 2$, where $m_2 = 1$ gives $(1 - t^3)^{-1}$. A single exact audit at five predeclared primes and four predeclared composites reproduces the theorem-controlled ledger with independent enumeration and formula engines. The collision $r_2 = r_4 = 3$ shows why the local exception supplies neither a common intrinsic modulus clock nor a prime selector. This is a scoped synthesis of established constructions, not a new zeta or stack theory and not a universal no-go statement.

Keywords: cat map, Burnside ring, equivariant zeta, finite permutation, orbifold reduction, quotient stack, periodic orbit.

1 Introduction and bounded question

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad R_q = \mathbb{Z}/q\mathbb{Z}, \quad q \geq 2. \quad (1)$$

A preceding finite-module audit identifies a canonical cyclic-vector locus $X_q = \text{CV}_q \subset R_q^2$ and proves that

$$G_q = \text{Cent}_{\text{GL}_2(R_q)}(A) = R_q[A]^\times \quad (2)$$

acts freely and transitively on X_q . The cat map acts on this torsor by left translation through the distinguished element $a_q = A \bmod q \in G_q$. The coarse quotient X_q/G_q is one point, and the induced transformation is identity because the transformation being studied already belongs to the group being divided out.

That one-point quotient poses a precise follow-up question. Can a standard equivariant, orbifold, or stack-level refinement retain enough of the erased motion to yield one intrinsic modulus-dependent local clock? A useful answer must distinguish several objects that are sometimes grouped under the phrase *equivariant zeta*. Fixed points, fixed group-orbits, full twisted fixed-point tables, enhanced equivariant sets, and quotient stacks retain different data and take values in different coefficient categories. The question is which information survives each named construction, what extra labels that retention costs, and whether any retained datum is a return-time law that identifies q or selects prime moduli.

The answer is a definition-sensitive hierarchy. The point-order rational Burnside series remembers source period. Its ordinary cardinality reduction also remembers the number of source cycles, while its additive orbifold reduction has a fractional exponent. The orbit-order integral Burnside series remembers the free orbit type but has only period one. Stronger G_q -permutation and enhanced carriers remember a_q because the regular action is effective and G_q is kept as a labelled ambient group. Morita-invariant quotient-stack data collapse to a fixed point. These are positive and negative boundaries of different constructions, not competing claims about one invariant.

Remark 1.1 (Post-run theorem-scope correction). Frozen design prose described the four scalar reductions with an overstrong per-row quantifier. The exact registered ledger contains one exception: at $q = 2$, $m_2 = 1$, so point-cardinality reduction is $(1 - t^3)^{-1}$ and has both source support and unit exponent. The registered control K011 checks the weaker and correct family statement: no single reduction type has both properties uniformly across all nine locked moduli. We use only that family-uniform statement. The raw formulas and results are unchanged, and the A0 conclusion below concerns the absence of a common intrinsic modulus or prime clock. A fresh semantic audit issued `PASS_WITH_SCOPE_CORRECTION` for this publication boundary.

This note makes five bounded contributions.

1. It gives one theorem for translation on an arbitrary finite abelian C -set, covering source cycles, coarse quotient cycles, both Burnside exact-period sequences, additive orbifold reductions, $\mathbb{Z} \times C$ stabilizers, action kernels, and quotient-stack inertia.
2. It separates additive exact-period reduction from Burnside-ring multiplication and supplies an explicit nonmultiplicativity witness.
3. It records effectivization and rigidification boundaries: the full twisted carrier recovers a only modulo the ineffective kernel, whereas isotropy that survives in BK components carries identity dynamics and disappears under componentwise rigidification.
4. It specializes the hierarchy to the regular cat-map centralizer torsor, deriving point-order, orbit-order, G -permutation, enhanced, orbifold, groupoid, shortening, and gluing records without conflating their coefficient categories.
5. It reports a source-locked exact audit on nine predeclared moduli and one separately typed C_6 structural control, with dual scientific engines and a separately reviewed post-run analyzer.

The contribution is deliberately modest. The underlying rational Burnside series, G -permutation invariant, integral orbit-order series, orbifold maps, enhanced carrier, and quotient-orbit mechanisms are prior art. We audit their interaction on one previously frozen torsor family. We do not define a new equivariant, Burnside, orbifold, groupoid, or stacky zeta; prove historical priority; construct a canonical comparison among the varying groups G_q ; or exclude unexamined representation-valued, weighted, analytic, transfer-operator, Fredholm, Hecke, quantized, Ruelle, or Fried refinements. No prime-zero correspondence or Riemann-hypothesis mechanism is asserted.

Section 2 fixes the prior-art and definition boundaries. Section 3 defines the finite carriers and additive reductions. Section 4 proves the general finite- C -set theorem. Section 5 treats effectivity, inertia, and rigidification, including the C_6 control. Section 6 specializes to the cat-map torsor. Section 7 reports the exact audit and provenance. Section 8 states the clock semantics, limitations, and route decision.

2 Prior art and definition boundary

Finite cat-map centralizers, reversing symmetries, rational lattices, and field-level normal types already have a direct dynamical treatment [2]. The regular G_q -torsor used here is imported from the separately

Table 1: Definition boundary for the audited constructions. Labelled means that the ambient finite group and its distinguished elements remain part of the coefficient data. No row is introduced as a new invariant.

Construction	Input used	Retained datum on a regular torsor	Main loss or cost
Point-order rational Burnside zeta	fixed-point classes $L^G(\phi^k)$	source order and regular orbit coefficient	rational exponent; untwisted data see the cyclic subgroup, not its chosen generator
Orbit-order integral Burnside zeta	fixed- G -orbit classes $\tilde{L}^G(\phi^k)$	regular orbit type	dynamical support only at period one
G -permutation zeta	all $L^G(g\phi^k)$	labelled twist modulo action kernel	a varying $\mathbb{Z} \times G$ -set category
Enhanced Burnside carrier	transformation and isotropy characters	labelled return twist	free torsor has only the trivial character
Orbifold/stack image	exact-period additive map or fixed sectors	static stabilizer sectors when present	no nontrivial quotient return period

finalized preceding audit. The present note neither reclassifies finite centralizers nor presents that torsor as a new rational-lattice theorem.

The rational point-order construction originates in the equivariant monodromy-zeta framework of Gusein-Zade et al. [5]. It uses equivariant Lefschetz classes in the rationalized Burnside ring together with the Burnside pre- λ /power structure. This source blocks any claim that the rational exact-period series below is a construction of the present note.

The strongest direct collision is the later treatment by Gusein-Zade et al. [6]. That work explicitly separates the equivariant Lefschetz class of fixed points from the alternative class of setwise fixed G -orbits. It defines the integral orbit-order zeta and two orbifold reductions, and its examples distinguish transformation order in a space from transformation order in a quotient. In particular, when the transformation belongs to the acting symmetry group, the induced quotient action is trivial. Our regular-torsor formulas are a finite specialization of that distinction.

A different invariant appears in Gusein-Zade [4]. It is a virtual finite or locally finite G -permutation, equivalently a $\mathbb{Z} \times G$ -set, determined by all twisted classes $L^G(g\phi^m)$. Its irreducible data include a stabilizer, a return integer, and a twist coset. This full twisted carrier is strictly stronger than the untwisted point-order series in the present example. The enhanced Burnside framework of Ebeling and Gusein-Zade [3] adds a transformation and isotropy characters and supplies a separate fixed-sector orbifold reduction. Again, we specialize those definitions rather than modify them.

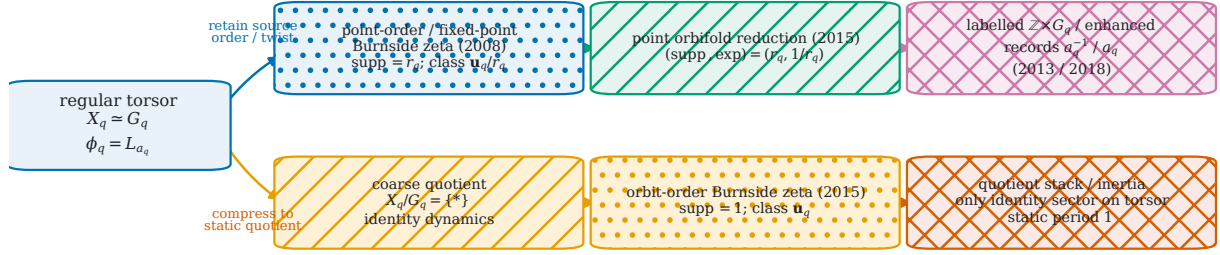
Quotient-orbit shortening and gluing follow the framework of Zegowitz [14]. Her orbit-intersection formulas identify exactly how many source cycles glue and by what factor a source period shortens. Miles [11] develops a different zeta for finitely generated acting groups through their finite-index subgroups and tacitly assumes an infinite acting group. It is a scope boundary, not an off-the-shelf finite- G_q replacement. Likewise, Walton [13] studies periodic points and twists on quotient varieties over finite fields; its hypotheses do not extend the present finite-set calculation to composite residue rings.

The current frontier uses equivariance and orbifolds in substantially different settings. Mackey-functor and cohomological zetas [12], cyclic-nerve symmetries [1], and equivariant Ruelle, trace, and Fried programs [8–10] do not share the finite Burnside definitions audited here. A recent survey also places monodromy-zeta variants in a mature literature [7]. These works prevent a narrow finite-set calculation from being presented as a general statement about every use of *equivariant* or *orbifold* zeta.

Table 1 records the nomenclature used throughout. Every row names a different carrier or reduction.

Figure 1 summarizes the definition firewall and the information retained by each named construction.

A named carriers retain different information



B information ledger on the regular torsor

carrier	order r_q	orbit type	labelled twist	isotropy	native period
point Burnside	yes	$[G/1]$	subgroup only	marks	r_q
point orbifold	yes	scalar	no	weight	r_q
orbit Burnside	no	$[G/1]$	no	marks	1
$\mathbb{Z} \times G$ perm.	yes	yes	a_q^{-1}	yes	labelled
enhanced carrier	yes	yes	a_q	character	labelled
stack / inertia	no	B1	no	identity	1

The rows are distinct established constructions; this is not a universal dominance order.

C scalar support/exponent ledger

reduction	support / exponent	scope
$\star \kappa(\text{point})$	support r_q , exponent m_q	unit when $m_q = 1$
$\Phi(\text{point})$	$(r_q, 1/r_q)$	fractional
$\kappa(\text{orbit})$	$(1, n_q)$	static support
$\Phi(\text{orbit})$	$(1, 1)$	static unit
$q = 2 \times \kappa(\text{point}) = (3, 1)$ some locked row type exception: $q = 4 \times \kappa = 3$, support 3 does not identify q .		

Family-level conclusion: no one scalar-reduction type combines source support and unit exponent uniformly over all nine locked rows.

Figure 1: Definition-sensitive information retention on the regular centralizer torsor. Panel A separates the point-order/fixed-point Burnside construction of 2008, the labelled $\mathbb{Z} \times G$ carrier of 2013, the orbit-order and orbifold constructions of 2015, and the enhanced carrier of 2018 [3–6]. Panel B records which information each named carrier retains; the rows are distinct established constructions, not a universal dominance order. Panel C gives the four frozen scalar reductions. In particular, point-cardinality has support r_q , exponent m_q , and unit exponent when $m_q = 1$. The star marks the sole locked exception $q = 2$, where $(r_2, m_2) = (3, 1)$; because $r_2 = r_4 = 3$, this coincidence is not modulus-specific. The certified negative statement is only family-uniform: no one scalar-reduction type combines source support and unit exponent over all nine locked rows.

3 Finite carriers and additive reductions

Let C be a finite abelian group, let X be a finite C -set, and let $\phi: X \rightarrow X$ be a C -equivariant bijection. Write $B(C) = K_0(\text{f. } C\text{-sets})$ for the Burnside ring. We use the Artin–Mazur inverse convention: a cycle of length d contributes $(1 - t^d)^{-1}$.

Definition 3.1 (Point-order exact classes). The fixed-point equivariant Lefschetz class is $L^C(\phi^k) = [\text{Fix}(\phi^k)] \in B(C)$. Its exact-period classes $P_m^C \in B(C)$ are defined by divisor inversion,

$$L^C(\phi^k) = \sum_{m|k} P_m^C. \quad (3)$$

The point-order rational Burnside zeta is

$$\zeta_{\text{pt}}^C(\phi; t) = \prod_{m \geq 1} (1 - t^m)^{-P_m^C/m}. \quad (4)$$

The coefficients in (4) generally lie in $B(C) \otimes \mathbb{Q}$. This is the fixed-point construction associated with the 2008 rational series; it is not the integral orbit-order construction.

Definition 3.2 (Orbit-order exact classes). Let $\tilde{L}^C(\phi^k)$ be the Burnside class of C -orbits that are setwise fixed by ϕ^k , decorated by their orbit types. Define $\tilde{P}_m^C$ by

$$\tilde{L}^C(\phi^k) = \sum_{m|k} \tilde{P}_m^C. \quad (5)$$

The orbit-order integral Burnside zeta is

$$\tilde{\zeta}_{\text{orbset}}^C(\phi; t) = \prod_{m \geq 1} (1 - t^m)^{-\tilde{P}_m^C/m}. \quad (6)$$

The divisibility theorem behind this construction makes (6) integral in the setting of Gusein-Zade et al. [6]. In our finite translation model the integrality will also be visible directly.

Definition 3.3 (Exact-period scalar reduction). For an additive map of abelian groups $\psi: B(C) \rightarrow R$, define

$$\psi_* \left(\prod_{m \geq 1} (1 - t^m)^{-S_m/m} \right) := \prod_{m \geq 1} (1 - t^m)^{-\psi(S_m)/m}. \quad (7)$$

The map is applied to exact-period exponent classes before the scalar product is formed.

Two maps will be used. Cardinality is $\kappa_C([Y]) = |Y|$. For abelian C , the orbifold map is the additive homomorphism

$$\Phi_C([C/K]) = |K|. \quad (8)$$

No multiplicativity of Φ_C is available or needed. Indeed, if $C \neq 1$ and $\mathbf{u} = [C/1]$, then the diagonal Cartesian product splits into $|C|$ regular orbits, so

$$\mathbf{u}^2 = |C|\mathbf{u}, \quad \Phi_C(\mathbf{u}^2) = |C| \neq 1 = \Phi_C(\mathbf{u})^2. \quad (9)$$

Equation (7) uses only the additive structure; it does not require ψ to preserve Burnside multiplication, the pre- λ structure, or the symmetric-power structure. Cardinality κ_C is in fact multiplicative on finite C -sets, whereas Φ_C fails multiplicativity by (9). For Φ_C , the definition records the 2015 procedure: reduce the Lefschetz sequence additively, perform divisor inversion, and form the scalar series.

The 2013 C -permutation invariant uses more input than either untwisted series. It is represented by the finite $\mathbb{Z} \times C$ -set obtained from

$$(j, c) \cdot x = c\phi^j(x). \quad (10)$$

Equivalently, it is determined by the complete table $L^C(c\phi^j)$. The enhanced carrier further keeps the transformation and isotropy characters. We reserve the names C -permutation zeta and enhanced Burnside carrier for these stronger labelled objects and do not identify them with (4) or (6).

4 Translation on a finite abelian C -set

Fix $a \in C$, put $H = \langle a \rangle$, and write the orbit-type decomposition

$$X \simeq \coprod_{K \leq C} n_K(C/K), \quad n_K \in \mathbb{Z}_{\geq 0}. \quad (11)$$

Because C is abelian, no conjugacy ambiguity remains in the subgroup index K . For each represented orbit type define

$$d_K = [H : H \cap K], \quad M_K = [C : HK]. \quad (12)$$

Theorem 4.1 (Finite translation hierarchy). *Let C be finite abelian and let $\phi_a(x) = ax$ act on (11). Then:*

1. *every point in a copy of C/K has exact period d_K , and that copy contains M_K translation cycles;*
2. *the ordinary source zeta and coarse quotient zeta are*

$$\zeta_{X, \phi_a}(t) = \prod_{K \leq C} (1 - t^{d_K})^{-n_K[C:HK]}, \quad (13)$$

$$\zeta_{X/C, \bar{\phi}_a}(t) = (1 - t)^{-\sum_K n_K}; \quad (14)$$

3. *the exact point-order and orbit-order Burnside classes are*

$$P_m^C = \sum_{d_K=m} n_K[C/K], \quad (15)$$

$$\tilde{P}_1^C = [X] = \sum_K n_K[C/K], \quad \tilde{P}_m^C = 0 \quad (m > 1); \quad (16)$$

4. *additive orbifold reduction gives*

$$\Phi_{C,*}(\zeta_{\text{pt}}^C) = \prod_K (1 - t^{d_K})^{-n_K|K|/d_K}, \quad (17)$$

$$\Phi_{C,*}(\tilde{\zeta}_{\text{orbset}}^C) = (1 - t)^{-\sum_K n_K|K|}; \quad (18)$$

5. *on C/K , the $\mathbb{Z} \times C$ -stabilizer under (10) is*

$$\widehat{K}_a = \langle \{0\} \times K, (1, a^{-1}) \rangle; \quad (19)$$

across all represented orbit types, this labelled carrier recovers a exactly modulo the action kernel

$$N = \bigcap_{n_K > 0} K; \quad (20)$$

6. *the quotient stack and its inertia decompose as*

$$[X/C] \simeq \coprod_K n_K BK, \quad (21)$$

with $\sum_K n_K|K|$ inertia components; translation by a induces identity dynamics up to 2-isomorphism on every component.

Proof. On C/K , the equality $a^j cK = cK$ is equivalent to $a^j \in K$. The least positive such j is the order of a in $H/(H \cap K)$, namely d_K . Every point has that period. Orbit-stabilizer then gives

$$\frac{[C : K]}{d_K} = \frac{|C| |H \cap K|}{|K| |H|} = \frac{|C|}{|HK|} = [C : HK] = M_K, \quad (22)$$

which proves the first claim and (13).

Each copy of C/K is one C -orbit. Since $a \in C$, translation is identity on the orbit quotient, proving (14). The points of exact period m are precisely the components with $d_K = m$, which proves (15). Every C -orbit is setwise fixed from the first iterate, which proves (16). Applying the additive map $\Phi_C([C/K]) = |K|$ to those exact-period classes, and only then forming the products, yields (17) and (18).

For the base coset K , a pair (j, c) stabilizes K precisely when $ca^j \in K$. This subgroup is generated by $\{0\} \times K$ and $(1, a^{-1})$, proving (19). It records the coset $a^{-1}K$. Two translation labels give the same recorded coset for every represented K exactly when they agree modulo the intersection (20).

Finally, the transitive action groupoid $C \ltimes (C/K)$ is equivalent to the one-object groupoid BK . Since K is abelian, its inertia is indexed by its $|K|$ elements. The arrow labelled a gives a natural isomorphism between the identity functor and translation by a ; naturality uses $ac = ca$. Thus each inertia component has static period-one dynamics. $\square$

Corollary 4.2 (What the untwisted series can identify). *The point-order sequence $\{P_m^C\}$ depends on a only through $H = \langle a \rangle$, so it cannot distinguish two generators of the same cyclic subgroup. The orbit-order sequence has no support beyond period one. The full labelled C -permutation carrier distinguishes a only modulo N , and distinguishes it exactly if and only if the C -action is effective.*

Remark 4.3 (Ordinary source zeta as cardinality reduction). Applying κ_C to (15) gives exponent $[C : K]/d_K = [C : HK]$ on each represented component. Hence the ordinary source zeta (13) is the point-order cardinality reduction. This identity does not turn Φ_C into a ring map and does not identify the stronger labelled carrier with either Burnside series.

5 Effectivity, inertia, and rigidification

The kernel N in (20) separates two operations that can otherwise look alike. First, *effectivization* replaces C by C/N . Each orbit becomes

$$C/K \simeq (C/N)/(K/N), \quad (23)$$

and the translation label becomes aN . The full twisted carrier recovers exactly this effective label. It does not canonically lift aN back to a chosen element of C .

Second, the quotient stack retains isotropy:

$$[X/(C/N)] \simeq \coprod_K n_K B(K/N).$$

The map from $[X/C]$ to this effective quotient removes the globally ineffective subgroup N ; in finite-groupoid language, it is the rigidification along the common ineffective stabilizer. Residual K/N -isotropy remains. If one then rigidifies each residual $B(K/N)$ component completely, that component becomes a point. Neither operation creates motion: translation was already 2-isomorphic to identity before effectivization, and every retained inertia component remains static.

This distinction gives three boundaries.

- Presentation-sensitive data such as the labelled $\mathbb{Z} \times C$ -set can retain aN .

- Morita-invariant quotient data can retain stabilizer groups or inertia multiplicities, but not a nontrivial translation period.
- Full rigidification removes the remaining static isotropy rather than converting it into a return clock.

No claim is made that every object called a stacky zeta factors through these operations. The statement concerns invariants that respect Morita equivalence and the 2-isomorphism class of the induced endomorphism.

5.1 The effective C_6 control

Let $C = C_6$, let a be a generator, and set

$$X = C_6/C_2 \sqcup C_6/C_3, \quad (24)$$

where C_j is the unique subgroup of order j . The action is effective because $C_2 \cap C_3 = 1$, so the labelled C -permutation carrier recovers a . Nevertheless,

$$d_{C_2} = 3, \quad d_{C_3} = 2, \quad M_{C_2} = M_{C_3} = 1.$$

The exact outputs are therefore

$$\zeta_{X, \phi_a}(t) = (1 - t^3)^{-1}(1 - t^2)^{-1}, \quad (25)$$

$$\zeta_{X/C, \bar{\phi}_a}(t) = (1 - t)^{-2}, \quad (26)$$

$$\Phi_{C,*}(\zeta_{\text{pt}}^C) = (1 - t^3)^{-2/3}(1 - t^2)^{-3/2}, \quad (27)$$

$$\Phi_{C,*}(\tilde{\zeta}_{\text{orbset}}^C) = (1 - t)^{-5}. \quad (28)$$

The quotient stack is $BC_2 \sqcup BC_3$, with five static inertia components. Thus effectivity and exact recovery of a labelled order-six translation do not force a period-six source factor. The C_6 object is a structural unit control; it is neither an arithmetic modulus row nor a second candidate.

Figure 3 below places this effective control beside the regular and trivial-action boundaries.

6 The regular cat-map centralizer torsor

We now specialize Theorem 4.1. The only imported arithmetic statement is the frozen terminal result of the preceding centralizer audit:

$$X_q = \text{CV}_q \simeq G_q \quad \text{as a left regular } G_q\text{-set}, \quad \phi_q(x) = a_q x, \quad a_q = A \bmod q. \quad (29)$$

No candidate or calculation from that centralizer-quotient audit is rerun here. Put

$$n_q = |G_q|, \quad r_q = \text{ord}_{G_q}(a_q) = \text{ord}_q(A), \quad m_q = \frac{n_q}{r_q}, \quad \mathbf{u}_q = [G_q/1] \in B(G_q). \quad (30)$$

In the notation of Section 4, the regular torsor has one orbit type $K = 1$, so $d_1 = r_q$ and $M_1 = m_q$.

Theorem 6.1 (Regular-torsor retention ledger). *For every $q \geq 2$, the following statements hold.*

1. *The source consists of m_q cycles of length r_q , and*

$$\zeta_{\phi_q}(t) = (1 - t^{r_q})^{-m_q}. \quad (31)$$

2. The point-order and orbit-order Burnside zetas are

$$\zeta_{\text{pt},q}^{G_q}(t) = (1 - t^{r_q})^{-\mathbf{u}_q/r_q}, \quad (32)$$

$$\tilde{\zeta}_{\text{orbset},q}^{G_q}(t) = (1 - t)^{-\mathbf{u}_q}. \quad (33)$$

3. Cardinality and additive orbifold reduction give exactly

$$\kappa_{G_q,*}(\zeta_{\text{pt},q}^{G_q}) = (1 - t^{r_q})^{-m_q}, \quad \Phi_{G_q,*}(\zeta_{\text{pt},q}^{G_q}) = (1 - t^{r_q})^{-1/r_q}, \quad (34)$$

$$\kappa_{G_q,*}(\tilde{\zeta}_{\text{orbset},q}^{G_q}) = (1 - t)^{-n_q}, \quad \Phi_{G_q,*}(\tilde{\zeta}_{\text{orbset},q}^{G_q}) = (1 - t)^{-1}. \quad (35)$$

4. The complete twisted table is

$$L^{G_q}(g\phi_q^k) = \begin{cases} \mathbf{u}_q, & g = a_q^{-k}, \\ 0, & g \neq a_q^{-k}. \end{cases} \quad (36)$$

Hence the G_q -permutation stabilizer is $\langle (1, a_q^{-1}) \rangle$, with irreducible triple $(\{e\}, 1, a_q^{-1})$.

5. The enhanced carrier is $\hat{X}_{\{e\},1,a_q,1}$, while its enhanced orbifold zeta is $(1 - t)^{-1}$.

6. The action groupoid $G_q \ltimes X_q$ is equivalent to one point, has no nonidentity inertia component, and carries an induced endofunctor naturally isomorphic to identity.

7. Every source cycle shortens by $1/r_q$, and precisely m_q source cycles glue to the quotient fixed point.

Proof. Under (29), $\phi_q^k(x) = a_q^k x$. Cancellation shows that one point is fixed if and only if $a_q^k = 1$, in which case every point is fixed. Thus every point has exact period r_q , giving (31) and

$$L^{G_q}(\phi_q^k) = \begin{cases} \mathbf{u}_q, & r_q \mid k, \\ 0, & r_q \nmid k. \end{cases} \quad (37)$$

Divisor inversion places the sole point exact class $\mathbf{u}_q$ at r_q , which proves (32). There is exactly one G_q -orbit and it is setwise fixed at every iterate. The sole orbit exact class is therefore $\mathbf{u}_q$ at support one, proving (33).

Regularity gives $\kappa_{G_q}(\mathbf{u}_q) = n_q$ and $\Phi_{G_q}(\mathbf{u}_q) = 1$. Applying these maps to the exact classes proves (34) and (35). A twisted iterate acts by translation through ga_q^k ; it fixes a point exactly when $g = a_q^{-k}$. This proves (36), the stabilizer, and the inverse convention in the G_q -permutation triple.

The regular action has trivial isotropy and one group orbit. The enhanced return twist is a_q , not a_q^{-1} , and its only character is trivial. Only the identity fixed sector is nonempty, so the enhanced orbifold image is one fixed point. The action groupoid is transitive with trivial automorphism groups. Arrows labelled a_q define a natural isomorphism from identity to the induced endofunctor. Finally, $|\mathcal{O}_{\phi_q}(x) \cap \mathcal{O}_{G_q}(x)| = r_q$, while $|\mathcal{O}_{G_q}(x)| = n_q$. The shortening and gluing values follow from the quotient-orbit formulas of Zegowitz [14]. $\square$

Proposition 6.2 (Corrected scalar quantifier). *The point-cardinality factor in (34) has source support and unit exponent exactly when $m_q = 1$. The point-orbifold factor has source support with exponent $1/r_q$, while both orbit factors have support one. In the locked nine-row ledger, the first condition occurs at $q = 2$ and nowhere else. Consequently, no one reduction type supplies source support and unit exponent uniformly across the locked family.*

Proof. The four support-exponent pairs read directly from (34)–(35):

$$(r_q, m_q), \quad (r_q, 1/r_q), \quad (1, n_q), \quad (1, 1). \quad (38)$$

The first pair has unit exponent precisely when $m_q = 1$. For the fixed matrix, $r_q > 1$ for every $q \geq 2$, because the off-diagonal entry of $A - I$ is 1. Thus $1/r_q \neq 1$, and support one is not source support. The exact ledger in Section 7 has $m_2 = 1$ and $m_q > 1$ for its other eight rows. $\square$

C21-corrected. No one of the four scalar-reduction types has source support and unit exponent for every modulus in the locked family. The unique locked row/type exception is $(q, j) = (2, \kappa\text{pt})$, where the source already consists of one cycle. We do not use the overstrong assertion that every row/type pair fails.

6.1 Labelled retention and coefficient cost

Equations (36) and the enhanced tuple give a positive retention statement: the stronger carriers recover the exact local translation a_q on the effective regular action. This is not a common clock across moduli. The values lie, respectively, in

$$K_0(\text{f. } G_q\text{-perm}) \quad \text{and} \quad \widehat{B}(G_q),$$

and the ambient labelled group G_q varies with q . The construction supplies no canonical cross- q identification. Reading the group name or the matrix label $A \bmod q$ as a modulus is therefore reading family metadata, not extracting a return time.

The quotient outputs make the opposite choice. They remove the labelled presentation and retain only a point with identity dynamics. On the regular torsor even isotropy disappears: the only inertia component is the identity sector. The labelled carriers and the Morita quotient are both valid, but they answer different questions.

7 Exact nine-row retention audit

The registered audit used exactly the ordered tuple

$$\mathcal{Q}_{\text{lock}} = (2, 3, 5, 7, 11, 4, 6, 9, 10) \tag{39}$$

and no other arithmetic modulus. The first five rows are predeclared prime controls and the last four are predeclared composite controls. The structural C_6 example in (24) occupied a separate namespace and never entered $\mathcal{Q}_{\text{lock}}$.

Table 2 reproduces exact integer and rational data from the registered result. The star is important: it is the unique locked row/type pair that combines source support with unit exponent.

Table 2: Exact support/exponent ledger. The point-cardinality reduction at $q = 2$ is $(3, 1)$, the sole locked row/type pair combining source support and unit exponent. No reduction type does so uniformly across all nine rows; the retained support still fails to identify the modulus because $r_2 = r_4 = 3$. Entries are exact and use the inverse Artin–Mazur sign convention.

q	n_q	r_q	m_q	$\kappa(\text{point})$	$\Phi(\text{point})$	$\kappa(\text{orbit})$	$\Phi(\text{orbit})$
(support, positive exponent)							
2	3	3	1	(3, 1)*	(3, 1/3)	(1, 3)	(1, 1)
3	8	4	2	(4, 2)	(4, 1/4)	(1, 8)	(1, 1)
5	20	10	2	(10, 2)	(10, 1/10)	(1, 20)	(1, 1)
7	48	8	6	(8, 6)	(8, 1/8)	(1, 48)	(1, 1)
11	100	5	20	(5, 20)	(5, 1/5)	(1, 100)	(1, 1)
4	12	3	4	(3, 4)	(3, 1/3)	(1, 12)	(1, 1)
6	24	12	2	(12, 2)	(12, 1/12)	(1, 24)	(1, 1)
9	72	12	6	(12, 6)	(12, 1/12)	(1, 72)	(1, 1)
10	60	30	2	(30, 2)	(30, 1/30)	(1, 60)	(1, 1)

Every row also has the following exact carrier data:

- the G_q -permutation twist is a_q^{-1} , and the enhanced return twist is a_q ;
- the effective regular action recovers the labelled a_q ;
- no nonidentity inertia sector is nonempty; and
- the coarse, enhanced-orbifold, and quotient-stack periods are one.

These statements live in separate raw-result namespaces. They were not derived by relabelling one scalar factor.

7.1 What the finite rows show

The rows expose two order collisions:

$$r_2 = r_4 = 3, \quad r_6 = r_9 = 12. \quad (40)$$

The first collision directly limits the $q = 2$ exception. Although $(1 - t^3)^{-1}$ is a one-cycle source factor, support three does not identify modulus two because composite modulus four has the same source order. The second collision shows the same noninjectivity at a larger period. All four composite controls satisfy the same retention hierarchy, so the construction has no intrinsic prime selector.

The family-level scalar control is recorded with its actual logical scope:

`K011 -- family-uniform source-support/unit-exponent nonattainment.`

Its first clause checks that not all point-cardinality exponents are one; it does not check that every such exponent is nonunit. The other clauses check that point-orbifold exponents are nonunit and that both orbit reductions have support different from r_q . Thus the stored Boolean is correct, while the stronger prose gloss preserved in some frozen design and review files is superseded by Remark 1.1 and Proposition 6.2.

Figure 2 gives the complete visual ledger and marks the sole locked exception rather than hiding it in a family aggregate.

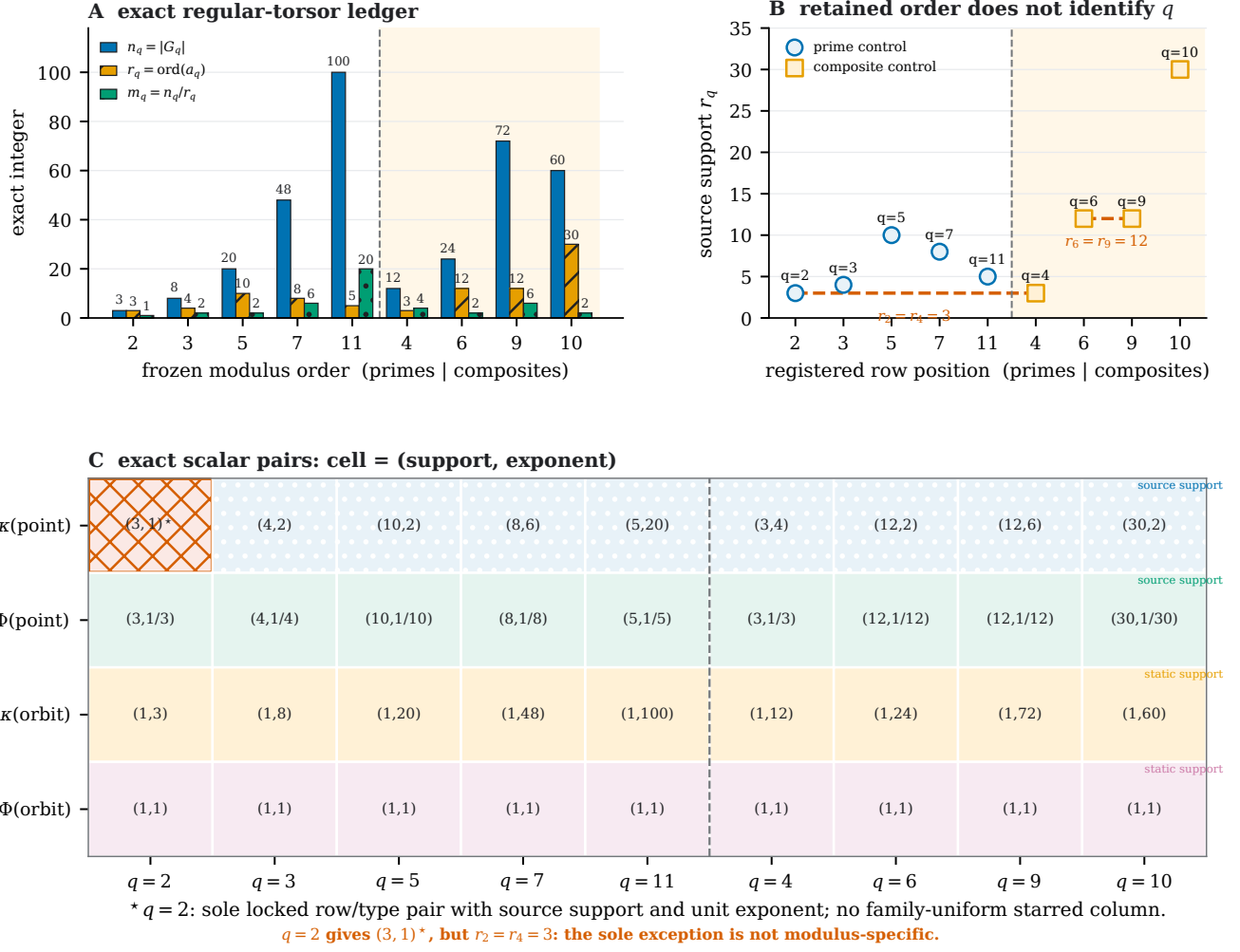
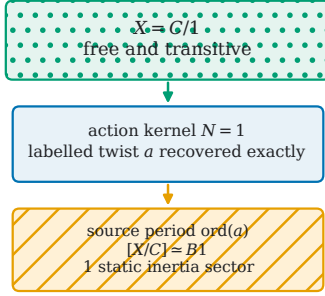
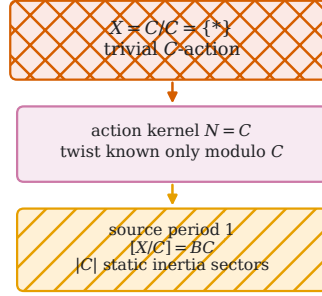


Figure 2: Exact nine-row retention ledger in the frozen order $q = (2, 3, 5, 7, 11, 4, 6, 9, 10)$. Panel A shows the integer triples (n_q, r_q, m_q) with $m_q = n_q/r_q$; panel B marks exactly the collisions $r_2 = r_4 = 3$ and $r_6 = r_9 = 12$; panel C lists all four scalar support–exponent pairs. The cell $(3, 1)^*$ at $q = 2$ is the sole locked row/type pair with source support and unit exponent. There is no family-uniform starred column, and $r_2 = r_4$ shows why the $q = 2$ exception is not modulus-specific. These finite rows are exact implementation and falsification controls, not a fit or an all- q inference.

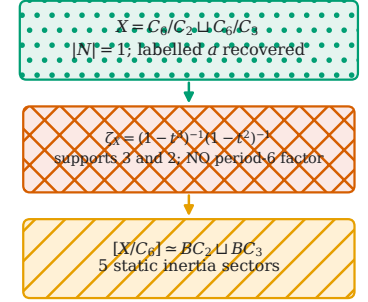
A regular orbit



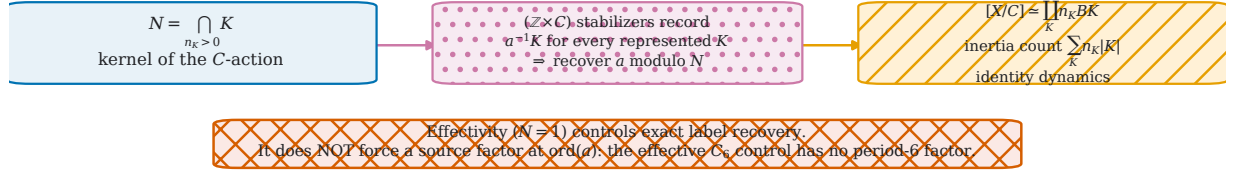
B trivial action / BC



C effective C_6 control



D what effectivity controls---and what it does not



The C_6 object is a structural control, not an arithmetic modulus row or a candidate; all inertia dynamics shown are static.

Figure 3: Effectivity controls labelled-twist recovery but does not turn quotient or inertia data into return times. Panel A is a regular orbit, for which the action kernel is trivial and the labelled twist is recovered exactly. Panel B is the trivial one-point action, whose quotient stack is BC_q with q static inertia sectors and whose twist is visible only modulo the full kernel. Panel C is the separately typed effective control $C_6/C_2 \sqcup C_6/C_3$: its labelled twist is recovered, while its source factors occur only at periods 3 and 2, never at period 6. Panel D states the general action-kernel and quotient-stack formulas. This structural control is not an arithmetic modulus row or a new candidate.

7.2 One registered run and two scientific engines

The audit was exact rather than statistical. It used no random seed, floating fit, external prime table, Riemann-zero data, numerical s , numerical $\log q$, or numerical q^{-s} . One engine explicitly enumerated residue matrices, cyclic vectors, group elements, source cycles, fixed sets, twisted fixers, and action-groupoid data. A separate engine constructed the same scientific records from the regular-torsor theorem. After removing only the engine identifier, their records agreed exactly.

A fresh verifier did not import or invoke the candidate. It independently enumerated the complete finite objects, reconstructed every formal factor and rational exponent, and matched all nine rows and the separate C_6 control. It also checked the general cyclic- C_n formulas on 264 subgroup fixtures with no discrepancy. This independent reconstruction issued `RESULT_PASS`. The finite audit validates transcription and implementation; the proof, not nine finite rows, remains the authority for the all- q statements.

7.3 Dual-tree closure and preserved failure history

The immutable registered execution tree and the post-run analyzer are separate authorities. The execution tree has digest

5ee1918a57fee56a2ca5a117c5749f614efbfd6baed96ae45480d6091a4741eb,

whereas the analyzer tree has digest

423082f4675a1d41622bcb3d090a2c4c67d4732ff6dc32d0298505d90d5a78c3.

The analyzer was permitted to validate immutable result artifacts but had no candidate authority.

The first manifest attempt failed before writing a file. Its K005 validator compared a JSON list with a Python tuple, so a correct unique-fixer record was rejected on type rather than scientific content. The raw result, claim, terminal, and execution tree remained unchanged. The separately reviewed analyzer repaired only this serialized-list boundary, reproduced all 174 unique-fixer checks, and then created the V2 manifest once. Read-only final closure passed, a second write was rejected, and the final result manifest has digest

a0b409061c34eff0d68fdc326fe4ec6ff9295895444b857ee161fd77e417292c.

This history matters because manifest closure establishes provenance and inventory integrity; it does not promote an imprecise prose quantifier into a theorem. The later semantic scope audit performs that separate task.

8 Clock semantics, limitations, and Route-A disposition

8.1 Retention is not a common return clock

The strongest audited carriers retain the local translation, but they do so inside a separately labelled G_q -category. The point-order series retains the integer r_q , but (40) shows that this integer is not a modulus identifier. The orbit-order, orbifold-compressed, and Morita quotient outputs have native period one. None of these facts supplies a common cross- q observable equal to q or $\log q$.

Substituting $t = q^{-s}$ imports the family label q . It is not generated by an iterate, fixed sector, groupoid automorphism, or source period. At $q = 2$, no exponent normalization is needed because the source already has one cycle. That local coincidence still requires the external substitution to become a q^{-s} factor, and its native support three also occurs at $q = 4$.

Although the $q = 2$ source already has one cycle and therefore yields the unit point-cardinality factor $(1 - t^3)^{-1}$, support three is not an intrinsic modulus clock. Across the locked family no scalar

reduction supplies a uniform source-support/unit-exponent construction, stronger coefficients remain in varying labelled categories, and composites obey the same hierarchy. The audited constructions therefore produce neither a common intrinsic modulus clock nor a prime selector.

8.2 Limits of the audit

The comparison is limited to named finite constructions:

1. the 2008 point-order rational Burnside zeta;
2. the 2015 orbit-order integral Burnside zeta and its two additive orbifold reductions;
3. the 2013 labelled G -permutation invariant;
4. the 2018 enhanced carrier and enhanced fixed-sector orbifold zeta; and
5. Morita- and 2-isomorphism-invariant data of the finite quotient groupoid.

The note does not compare every representation-valued, weighted, derived, cohomological, analytic, or presentation-sensitive refinement. It does not extend the acting-group zeta of Miles [11] to finite G_q , extend the finite-field quotient results of Walton [13] to composite residue rings, or enter the analytic Ruelle/Fried programs cited in Section 2.

The general theorem assumes C abelian. This is sufficient for the centralizer groups G_q and makes the BK inertia count equal to $|K|$. Nonabelian groups would require conjugacy-sensitive orbit types, centralizers, and naturality conventions. Nothing here claims that the same formulas transfer unchanged.

The exact audit is development-seen and finite. It is a falsification and implementation control, not statistical evidence and not a proof by exhaustion of all moduli. Its values were selected before the registered execution, inherited from the preceding audit, and include composites precisely to test non-prime specificity.

8.3 Formal route decision

Within this frozen scope, the result supports

```
EQUIVARIANT_RETENTION_COMPRESSION_TRADEOFF_CERTIFIED,  
A0_FAIL_MODULUS_GLOBAL_NON_SPECIFIC,    ROUTE_B_NOT_OPENED.
```

The first token means family-nonuniform retention/compression, with the $q = 2$ exception stated above; it does not mean pointwise mutual exclusion at every modulus. The A0 token records failure to obtain a common intrinsic modulus or prime clock, not failure of every local one-cycle factor. Route B remains unopened: no transfer, Fredholm, Hecke, quantization, Ruelle/Fried, prime-zero, or analytic claim follows from this finite hierarchy.

9 Conclusion

Standard finite equivariant constructions do retain information erased by a coarse centralizer quotient, but they retain different information. Fixed points retain source order, fixed group-orbits retain a static orbit type, and full twisted or enhanced carriers retain a translation label modulo the action kernel. Orbifold and Morita quotient images compress the regular torsor to identity dynamics.

The exact cat-map specialization makes the cost visible. A labelled local carrier varies with q ; an unlabelled quotient loses the clock; and the retained order collides across prime and composite moduli. The exceptional one-cycle factor at $q = 2$ is real and explicit, but it has support three, which also occurs

at $q = 4$. The audited standard constructions therefore do not furnish a common modulus clock or prime selector.

This low-novelty boundary note closes only that named Route-A branch. Its value is the definition separation, scope correction, and exact provenance, not a claim to a new equivariant or stacky zeta. Stronger analytic or globally comparable mechanisms would require a new source lock and lie outside the present result.

A Divisor inversion and exact-period details

For completeness, let $F_k \in B(C)$ be a sequence with $F_k = \sum_{m|k} S_m$. Möbius inversion gives

$$S_m = \sum_{d|m} \mu(m/d) F_d. \quad (41)$$

On a component C/K , $F_k = [C/K]$ precisely when $d_K \mid k$, and is zero otherwise. Equation (41) therefore yields one exact point class $[C/K]$ at d_K . Summing the n_K copies gives (15).

For fixed C -orbits, the class is $[C/K]$ at every iterate because translation stays inside the unique orbit. Möbius inversion then places $[C/K]$ only at support one. Summing components proves (16). This direct argument also shows why the point-order factor can be rational while the orbit-order factor here is integral.

B Twists, kernels, and groupoid naturality

The stabilizer calculation in (19) keeps track of the inverse convention. Under the locked left action $(j, c) \cdot xK = ca^j xK$, the base coset is fixed when $ca^j \in K$, so the positive generator is $(1, a^{-1})$. The enhanced carrier instead records the return transformation itself, namely a . These appearances of a^{-1} and a are compatible and must not be interchanged.

If $a, b \in C$, their stabilizer data agree on C/K exactly when $aK = bK$. Agreement across all represented K is therefore equivalent to $aN = bN$, with N from (20). This proves exact recovery modulo the action kernel without asserting an intrinsically chosen generator after the ambient group label is forgotten.

For the action groupoid, let $F_a(x) = ax$. At every object x , choose the arrow $\eta_x: x \rightarrow ax$ labelled a . An arrow labelled c gives the naturality square

$$c \circ a = a \circ c,$$

which commutes because C is abelian. Thus $\eta: \text{Id} \Rightarrow F_a$ is a natural isomorphism. Any invariant depending only on the Morita class of the groupoid and the induced endomorphism up to 2-isomorphism must assign this pair its identity value. Presentation-sensitive Burnside and permutation carriers are not covered by that conditional statement.

C Evidence and lifecycle bindings

Table 3 lists the principal immutable authorities used by the manuscript. Publication assets entered only after the independent ASSET_PASS; manuscript-source and build hashes are bound separately in the accompanying pre-review integrity record.

Table 3: Core evidence bindings. Digests are SHA-256.

Authority	Digest / disposition
source lock v2	331a1f9004f83c7979daf8eacddd6844072c6b5b7068293c1276985cf6aaa87b
frozen upstream centralizer-audit PDF	f685996c741c3e92d4eb18086f2a4e4d898ede10e8124a23991ada3579f8d378; terminal complete
independent source rereview	2f75d6934e3d61bdc941ee6689102a1cb08a959270a7cd87965579f1ec5cc622; SOURCE_LOCK_PASS
frozen proof/formula package	3d723fdb02c89f9b2f281da807bcd745c5991393d25e223f95d6673961c20948; scalar quantifier superseded by the scope audit below
immutable execution tree	5ee1918a57fee56a2ca5a117c5749f614efbfd6baed96ae45480d6091a4741eb
raw exact result	bef8aa5d632ed11b1ca58a123bbfe967a5426e2049d862118a373e4c1dc005fe
independent result integrity	c91737c8bf860bd559eebebe08420fc5d095800c47d132381f584e918e714a20; RESULT_PASS
post-run analyzer tree	423082f4675a1d41622bcb3d090a2c4c67d4732ff6dc32d0298505d90d5a78c3
independent analyzer review	ba63afc8c88903f15ec6ac5d82f0cd65430710ca9c132b489a7cd4f70e7660a8; ANALYZER_PASS
dual-tree result manifest	a0b409061c34eff0d68fdc326fe4ec6ff9295895444b857ee161fd77e417292c; final read-only closure pass
independent theorem-scope audit	f7b365c9e6c8933cf3cbcaf3c96692cbacdaabcc84400bdc629f1d482cb243e4; PASS_WITH_SCOPE_CORRECTION
frozen publication asset tree	95bb23519a427ef6a73a6a04b1aef6861aa4c5e4f6b844e7866bf9c43e52b28c
frozen figure manifest	e3a8d1d36ba8c4959b080a9661b242c40195ea08d27690fc8ee899b487cfd6dc
independent plan/figure review	ebf1644dc03da4c1ccc03972b545688d595ed6da125de2ec831ffc82e4e69cf; ASSET_PASS

The registered command ran once. The manuscript stage did not rerun the candidate, either test suite, the independent scientific verifier, or the post-run analyzer. It did not alter any file under `code/`, `experiments/`, or `results/`.

D Bibliographic verification boundary

The bibliography uses primary or official metadata records frozen by the publication asset gate. The 2008, 2013, 2015, and 2018 records are direct construction authorities. Zegowitz supplies shortening/gluing terminology; Miles and Walton are scope boundaries only. Current cohomological, cyclic-nerve, Ruelle, trace, Fried, and survey records prevent unsupported generality or priority language.

The publication layer corrects one historical sidecar transcription. The DOI-authoritative record for Walton’s paper is *Journal of Number Theory* **192** (2018), 386–405, DOI 10.1016/j.jnt.2018.03.023. The frozen design sidecar records volume 189, pp. 202–223; it remains immutable as provenance. The corrected metadata are used only in this bibliography and change no source formula, theorem, result, novelty conclusion, or scientific scope.

Availability and declarations

Data and code availability. The accompanying local research package contains the source lock, proof package, reviewed exact-arithmetic implementation, single registered result, independent reviews, theorem-scope correction, and strict result manifest. No external dataset is used by the mathematical claims or finite audit.

Ethics declaration. The work involves no human participants, animals, personal data, or intervention study.

Author contributions. Authorship and CRediT-role metadata are withheld in this anonymous pre-review copy and will be supplied for any non-anonymous release.

Conflict of interest. Conflict-of-interest metadata are withheld with the anonymous author record and must be completed before external submission.

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AI-assisted workflow disclosure. AI-assisted research and writing tools were used to organize source-locked materials, draft text, and support mechanical checks. Mathematical claims, citations, finite outputs, theorem-scope corrections, and lifecycle statements were separately bound to recorded source, code, result, and review gates. Responsibility for released content remains with the authors.

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